Compact Particle Launcher for Investigating Particle–Fluid Interface Interactions at the Capillary Scale

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Senior Honors Thesis
Sc.B., Mechanical Engineering
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Engineering Research Honors Thesis Symposium
Brown University School of Engineering
Providence, RI
USA
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Agenda

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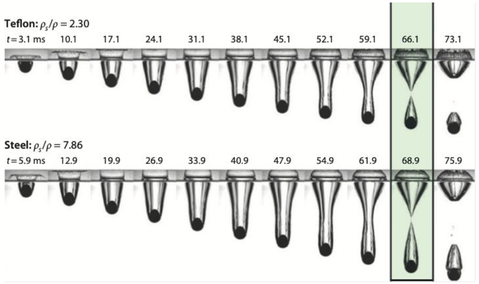
Particle-Fluid Interface Interactions

Particle-Planar Interface Impact Interactions



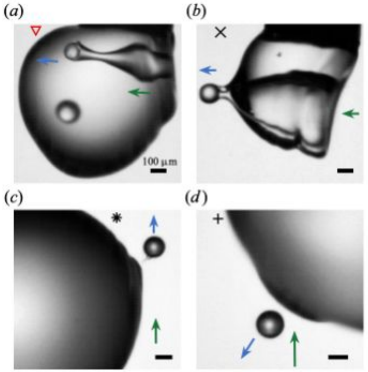
Cavity pinch-off. T. T. Truscott et al. (2014).

Cavity formation. C. Duez et al. (2007).

Implications:

- Autonomous underwater vehicle design
- Space capsule design

Particle-Droplet Impact Interactions



Droplet capture, pass-through, rebound. N. B. Speirs et al. (2023).

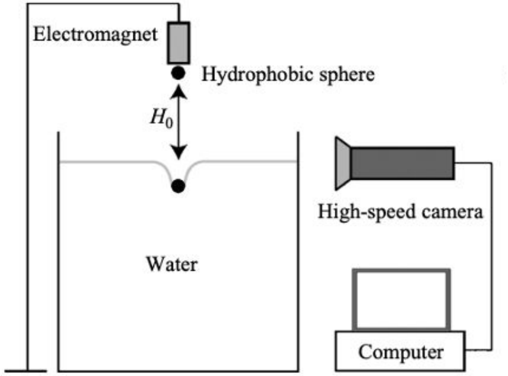
Implications:

- Raindrop particulate pollutant capture
- Wet scrubber systems for gaseous control

Particle-Fluid Interface Interactions

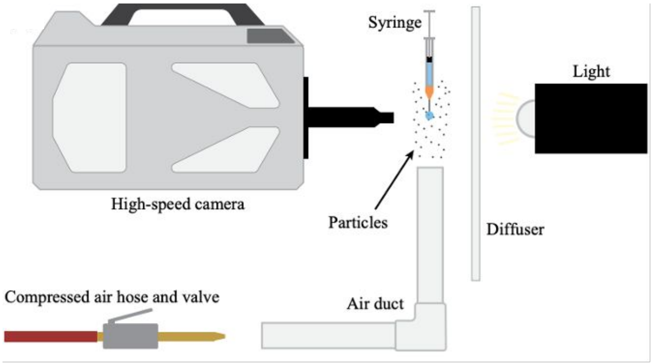
Existing experimental methodology & apparatus

Flat interface impact:



Water entry apparatus. J. M. Aristoff et al. (2009).

Droplet impact:



Droplet impact. N. B. Speirs et al. (2023).

Limitations

Flat interface:

- Sphere material
- Terminal velocity
- Confined space

Droplet:

- No precise velocity control
- No single-impact interactions
- Compressed air may affect interactions

Research Gap

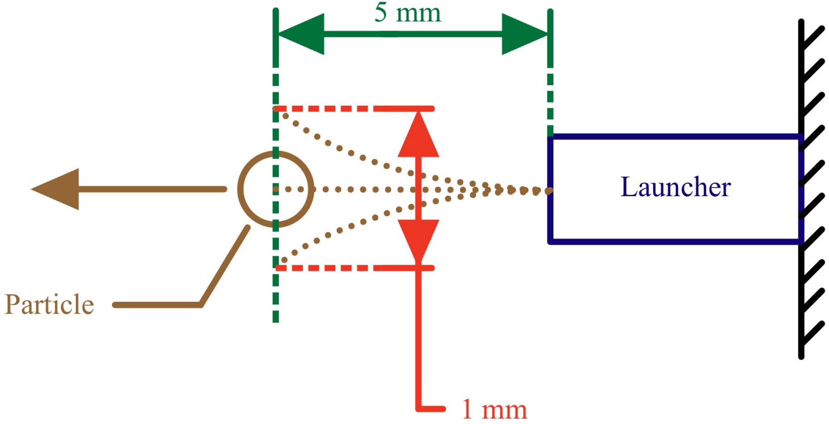
Limited existing experimental methodologies prohibit:

- *Water entry in vacuum chamber*
    - Study of effects of ambient pressure on cavity formation at capillary scale
- *Single-impact droplet experiments*
- Low density particle impacts
- High velocity impact experiments
- Millimetric scale particle impacts

→A compact device to precisely accelerate individual millimeter-scale particles to specific target velocities could allow this research gap to be filled.

Key Design & Performance Requirements

1. **Confined space operation**
   a. 244 x 244 x 310 mm test section (vacuum chamber)
   b. Inaccessible when sealed
2. **Particle size: 1-2 mm diameter spheres**
   a. Millimetric scale; N. B. Speirs et al. (2023).
3. **Target launch velocity range: 0.9-5 m/s**
   a. ±0.25 m/s uncertainty
   b. Span range from prior work
4. **Trajectory deviation threshold**
   a. ≤0.5 mm deviation @ 5 mm travel
   b. Ensures particle does not miss droplet



*Not to scale

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Iterative Design Process: Early Versions

Spring-based propulsion


Version **1**


Version **2**


Version **3**

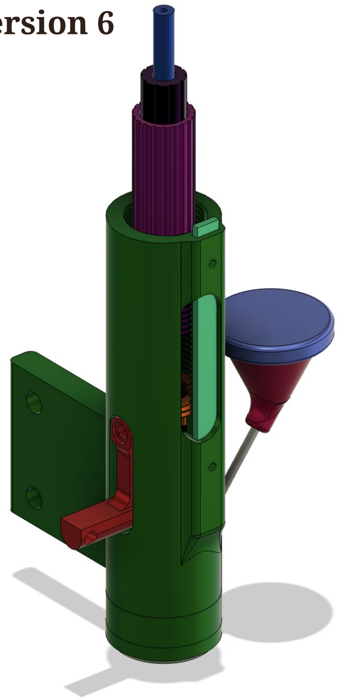

Version **4**

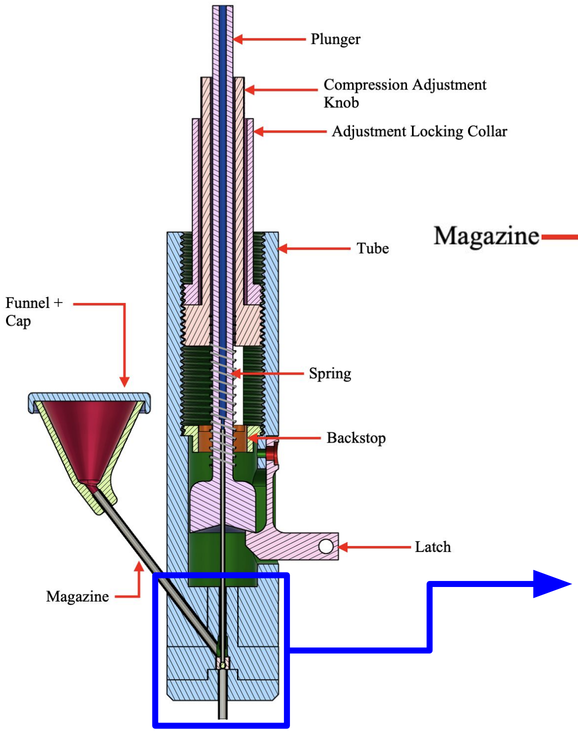

Version **5**

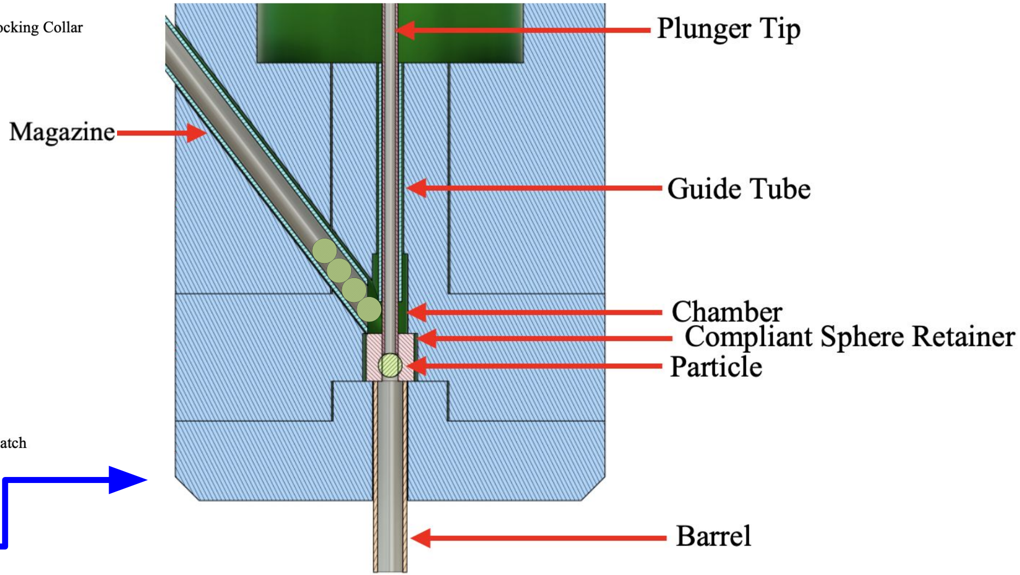
Plunger impact — Compliant sabot release — Hypodermic tube-guided launch

Final Launcher Design Overview

Version 6







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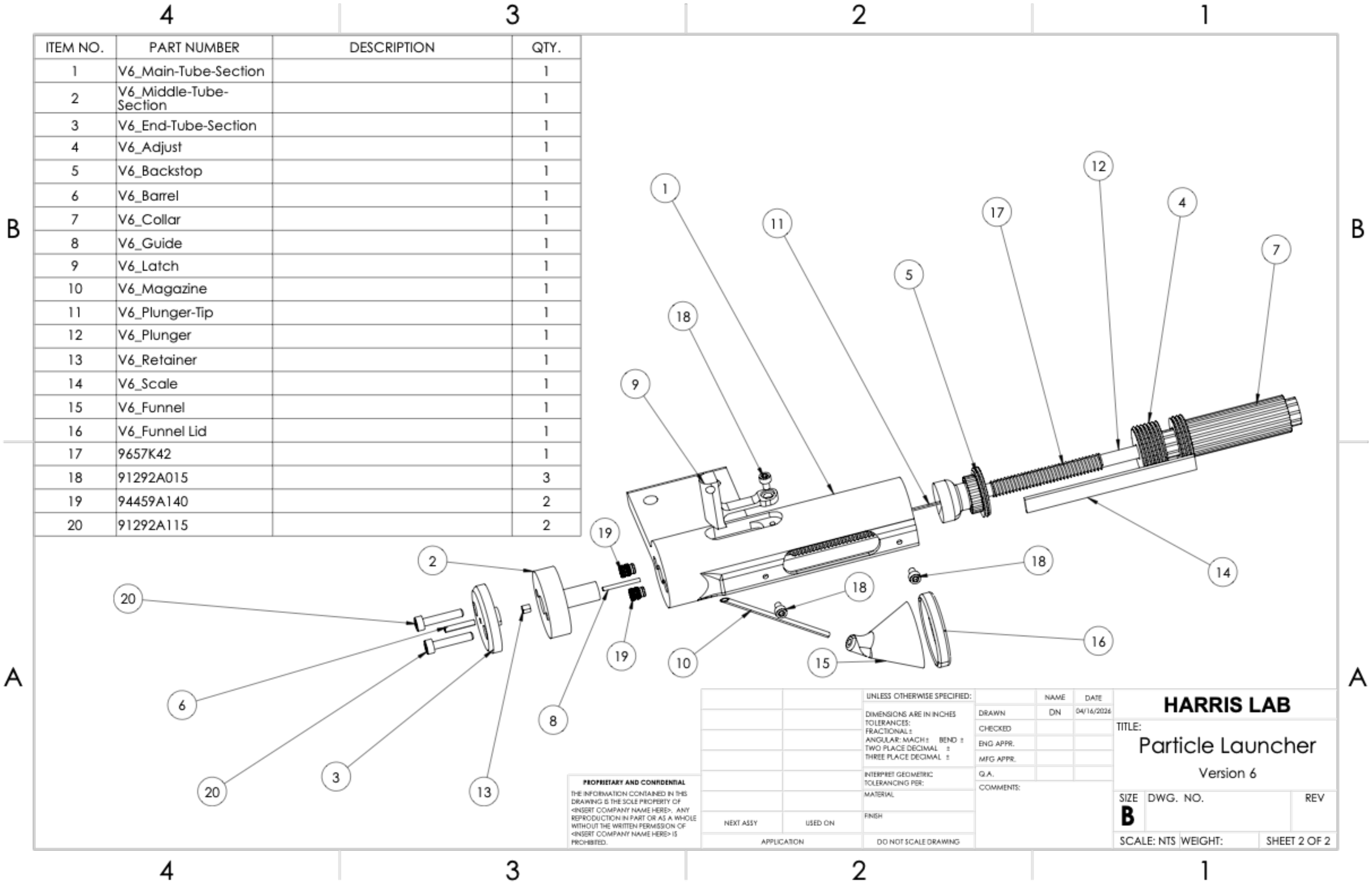
Final Launcher Design Operation

Final Launcher Design Overview



ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	V6_Main-Tube-Section		1
2	V6_Middle-Tube-Section		1
3	V6_End-Tube-Section		1
4	V6_Adjust		1
5	V6_Backstop		1
6	V6_Barrel		1
7	V6_Collar		1
8	V6_Guide		1
9	V6_Latch		1
10	V6_Magazine		1
11	V6_Plunger-Tip		1
12	V6_Plunger		1
13	V6_Retainer		1
14	V6_Scale		1
15	V6_Funnel		1
16	V6_Funnel Lid		1
17	9657K42		1
18	91292A015		3
19	94459A140		2
20	91292A115		2

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Final Launcher Design Overview

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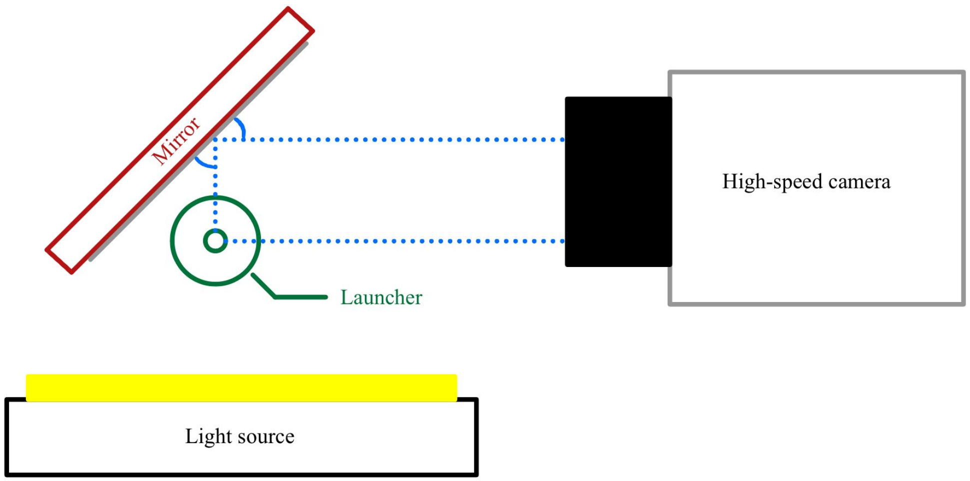
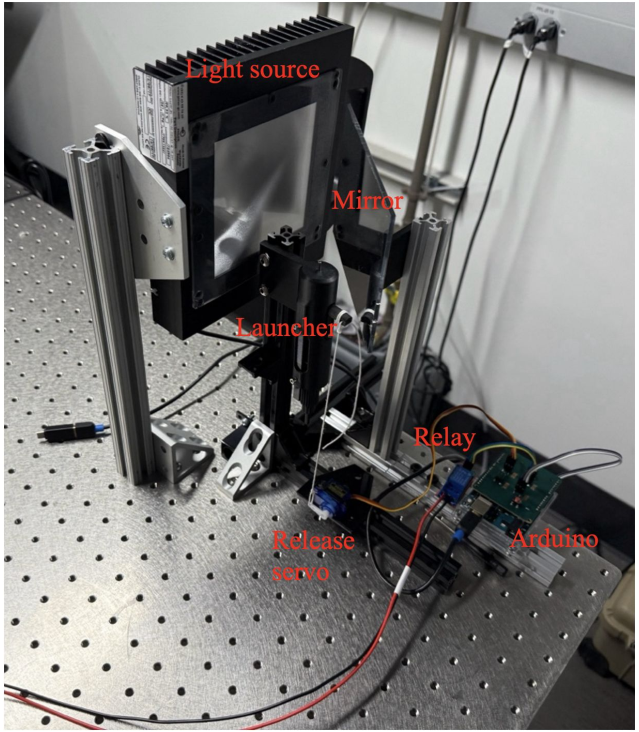
Performance Evaluation Methodology

Apparatus





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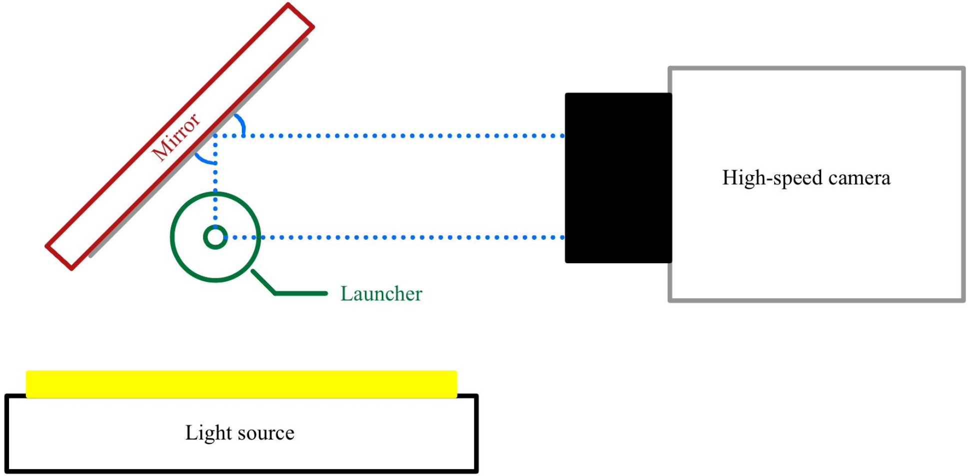
Performance Evaluation Methodology

Apparatus



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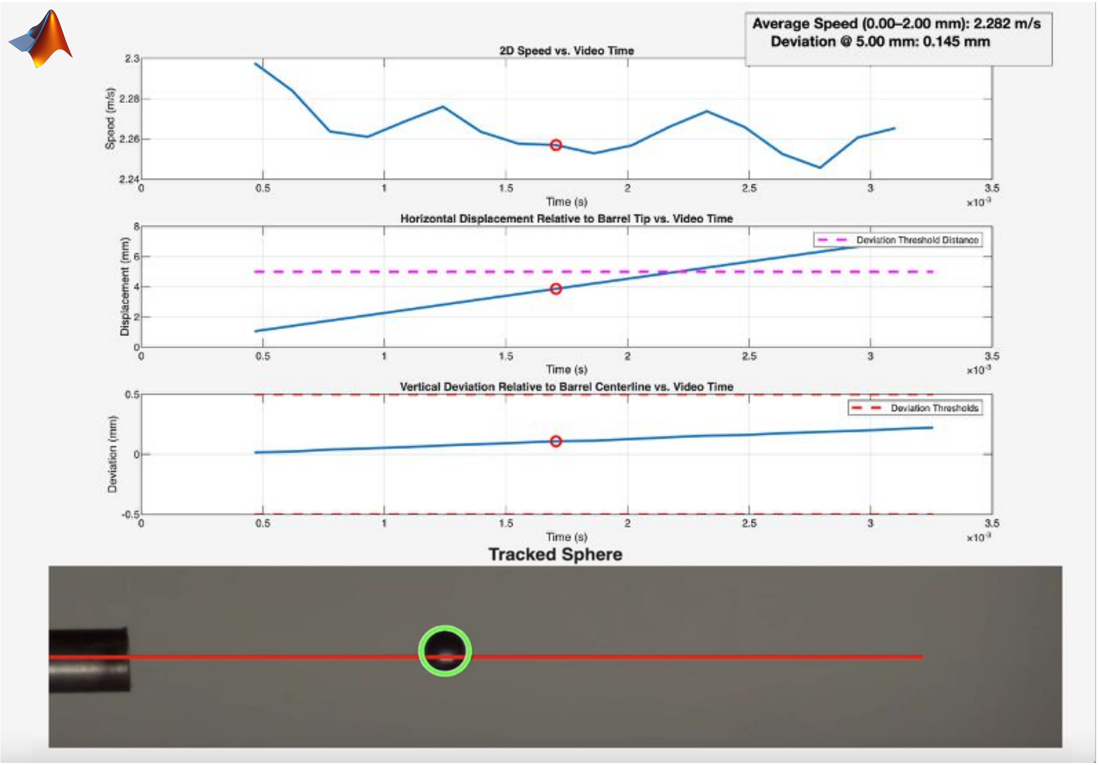
Performance Evaluation Methodology

Trajectory Measurement (MATLAB Image Processing)



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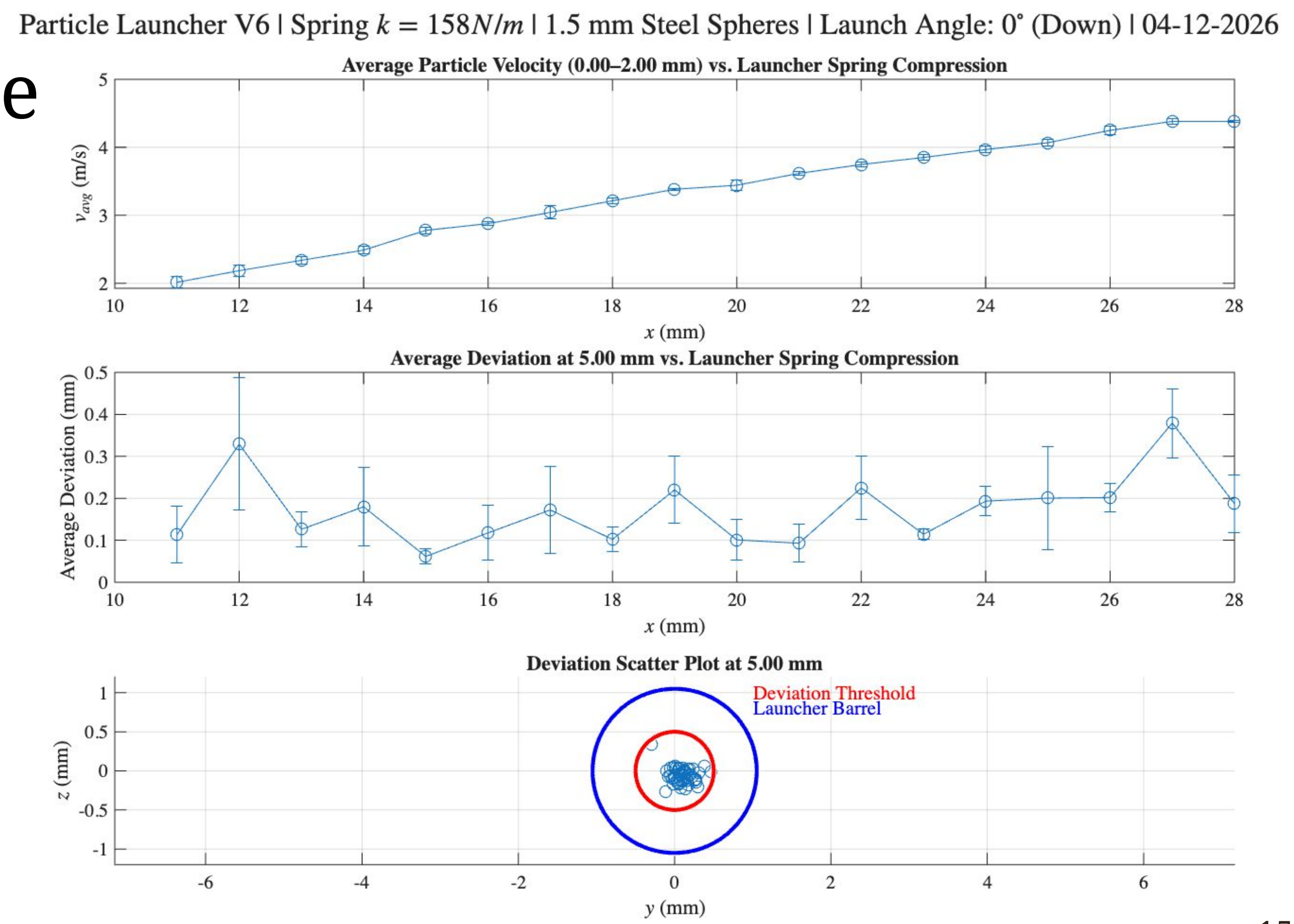
Performance Results



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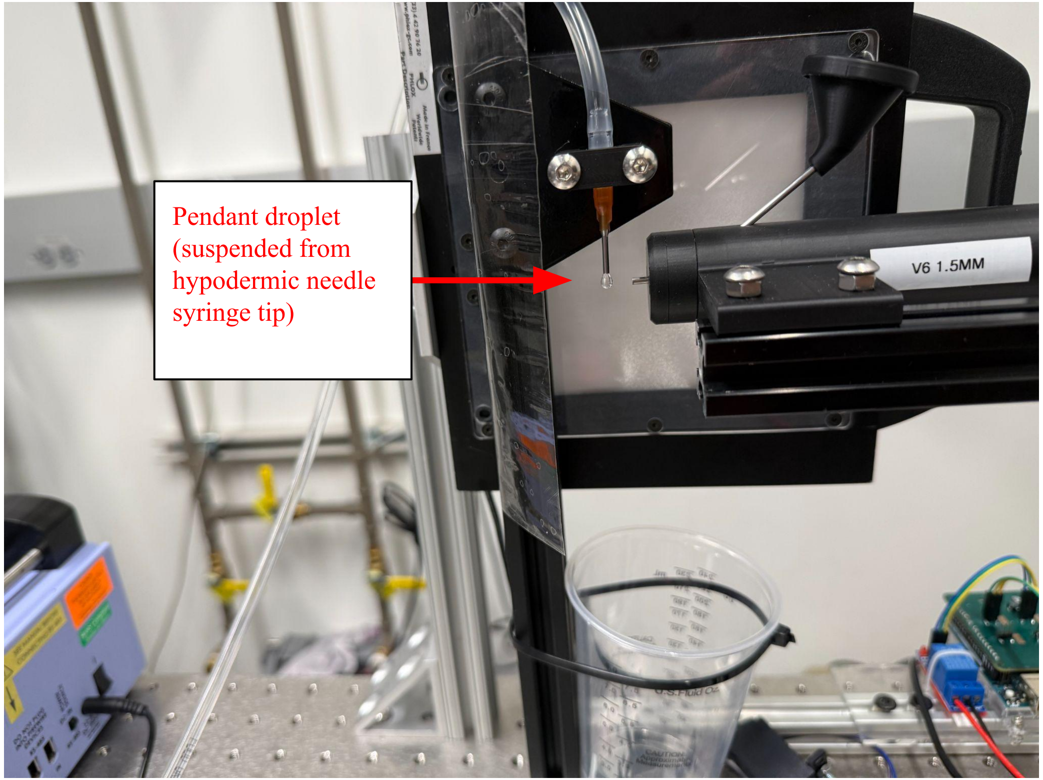
Sample Experimental Performance

Apparatus

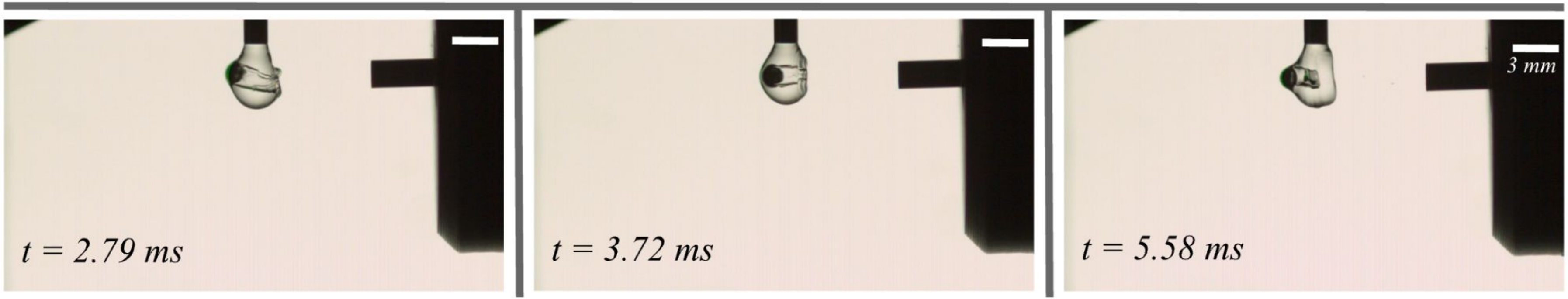


Sample Experimental Performance

Sample Experimental Performance

Pendant droplet impacts at varying velocities

$v_{launch} \approx 3.7\ m/s$ $v_{launch} \approx 2.6\ m/s$ $v_{launch} \approx 2.3\ m/s$



Conclusions

Launcher Evaluation

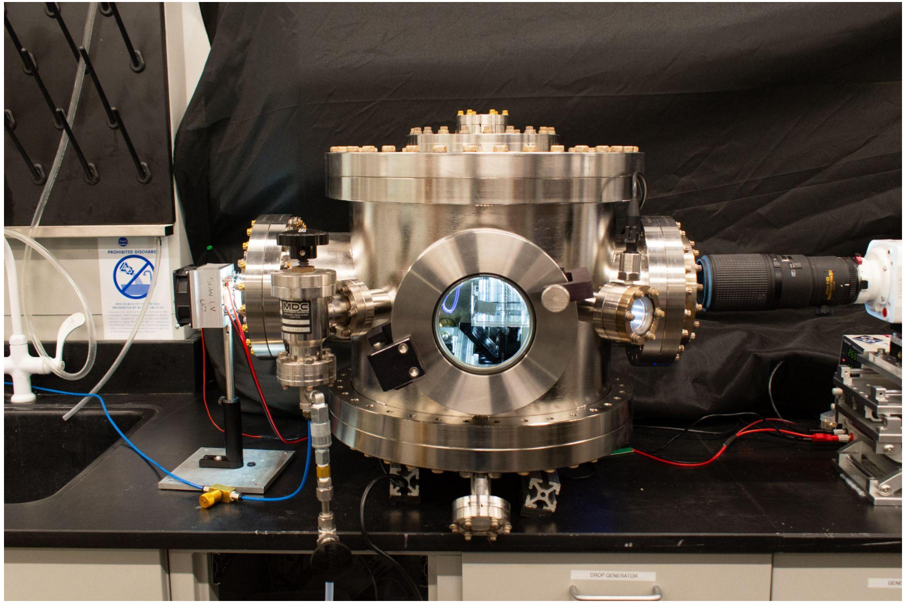
- Operates & performs as required
- Fulfills design requirements
- Shown to be effective in droplet impact experiments

Limitations

- Lack of exhaustive testing
  - Launch data
  - Fatigue & failure testing
- Sphere size and velocity

Future Work

- Launcher improvements, additional features
- Theory refinement
- Use of launcher for interfacial impact experiments!



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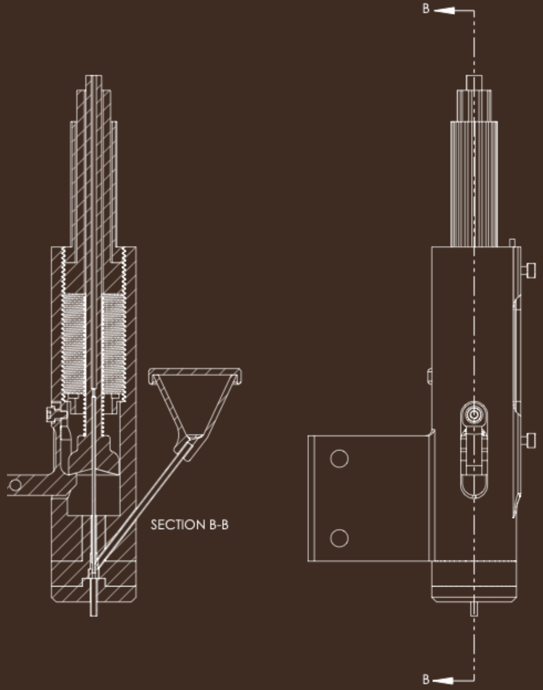
Eli Silver *Research Engineer*

Harris Lab

Engineering Honors Program Committee

Office of Naval Research (ONR N00014-21-1-2816)

Thank You!



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References

Aristoff, Jeffrey M., and John W.M. Bush. 2009. "Water entry of small hydrophobic spheres." Journal of Fluid Mechanics 619: 45-78.

Christie, Dan Edwin. 1952. Intermediate College Mechanics: A Vectorial Treatment. New York: McGraw-Hill Book Company, Inc.

n.d. Contact Angle Measurements and Wettability. Nanoscience Instruments. Accessed April 16, 2026. https://www.nanoscience.com/techniques/tensiometry/contact-angle-measurements-and-wettability/.

Crupano, Walter, Barbara Adrover-Montserrat, Jordi Lluma, Ramon Jerez-Mesa, and J Antonio Travieso-Rodriguez. 2024. "Investigating mechanical properties of 3D printed polylactic acid / poly-3-hydroxybutyrate composites. Compressive and fatigue performance." Heliyon 10 (18).

Drysdale, William H, Richard D Kirkendall, and Louise D Kokinakis. 1978. Sabot Design for a 105mm APFSDS Kinetic Energy Projectile. Aberdeen, MD: US Army Ballistic Research Center.

Duez, Cyril, Christophe Ybert, Christophe Clanet, and Lydéric Bocquet. 2007. "Making a Splash with Water Repellency." Nature Physics 3: 180-183.

Eshraghi, Javad, Sunghwan Jung, and Pavlos P Vlachos. 2020. "To seal or not to seal: The closure dynamics of a splash curtain." Physical Review Fluids 5.

Galeano-Rios, Carlos A, Radu Cimpeanu, Isabelle A Bauman, Annika MacEwen, Paul A Milewski, and Daniel M Harris. 2021. "Capillary-scale solid rebounds: experiments, modelling and simulations." Journal of Fluid Mechanics 912: A17-1 - A17-31.

Gilbarg, David, and Robert A Anderson. 1948. "Influence of Atmospheric Pressure on the Phenomena Accompanying the Entry of Spheres into Water." Journal of Applied Physics 19 (2): 127-139.

Huang, Zhen-gui, Mahfouz Elnaggar Wessam, and Zhi-hua Chen. 2014. "Numerical investigation of the three-dimensional dynamic process of sabot discard." Journal of Mechanical Science and Technology 28 (7): 2643.

Hurd, Randy C, Jesse Belden, Michael A Jandron, D Tate Fanning, Allan F Bower, and Tadd T Truscott. 2017. "Water entry of deformable spheres." Journal of Fluid Mechanics 824: 912-930.

n.d. Miniature Stainless Steel Tubing. McMaster-Carr. Accessed April 12, 2026. https://www.mcmaster.com/products/miniature-stainless-steel-tubing/miniature-stainless-steel-tubing-7/.

2025. Monitoring by Control Technique - Wet Scrubber For Gaseous Control. United States Environmental Protection Agency. October 20. Accessed April 13, 2026. https://www.epa.gov/air-emissions-monitoring-knowledge-base/monitoring-control-technique-wet-scrubber-gaseous-control.

Oberg, Erik, Franklin D Jones, Holbrook L Horton, and Henry H Ryffel. 2004. Machinery's Handbook. Vol. 27th Edition. New York, NY: Industrial Press, Inc.

2024. Parafrog Airsoft. September 20. Accessed April 12, 2026. https://parafrogairsoft.com/blogs/news/how-do-airsoft-guns-work.

n.d. "PLA Basic - Technical Datasheet V3.0." Bambu Lab. Accessed April 15, 2026. https://wiki.bambulab.com/filament-acc/abs-asa-pc/bambu_pla_basic_technical_data_sheet.pdf.

Prasad, B Sairam, G. Ravi Kiran Sastry, and H.N. Das. 2024. "A comprehensive review study on multiphase analysis of water entry bodies." Ocean Engineering 292.

Ray, Pradosh K. 1984. ARC-DRIVEN RAIL GUN RESEARCH. Tuskeegee, AL: NASA. https://ntrs.nasa.gov/api/citations/19850013352/downloads/19850013352.pdf.

Speirs, Nathan B, Jesse L Belden, and Aren M Hellum. 2023. "The capture of airborne particulates by rain." Journal of Fluid Mechanics 958.

Travieso-Rodriguez, J Antonio, Ramon Jerez-Mesa, Jordi Llumá, Oriol Traver-Ramos, Giovanni Gomez-Gras, and Joan Josep Roa Rovira. 2019. "Mechanical Properties of 3D-Printing Polylactic Acid Parts subjected to Bending Stress and Fatigue Testing." Materials (MDPI) 12.

Truscott, Tadd T, Brenden P Epps, and Jesse Belden. 2014. "Water Entry of Projectiles." Annual Review of Fluid Mechanics 46: 355-378.

von Kármán, Theodore. 1929. The impact on seaplane floats during landing. Washington, D.C.: National Advisory Committee for Aeronautics.

Appendix

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Particle-Planar Interface Interactions

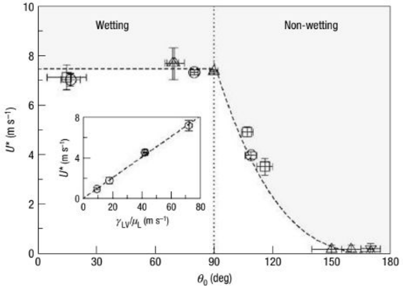
When a solid particle (sphere) impacts flat gas-liquid interface (surface of water body)...

- Effect of fluid on solid (impact forces)
- *Effect of solid on fluid*
    - Gas cavity
        - Formation, size/depth, closure/pinch-off

Controlling Parameters

- Particle entry velocity, size, material properties, wettability
- Liquid properties
- Ambient gas properties (pressure)

C. Duez et al. (2007)



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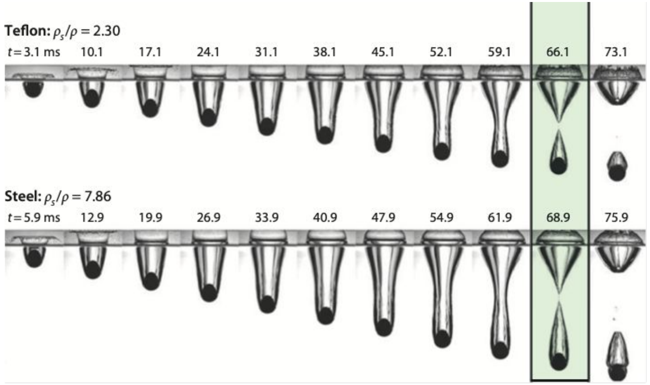


T. T. Truscott et al. (2014)

Implications

- Biological systems
- Autonomous underwater vehicle design
- Space capsule design

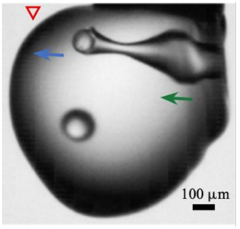
Particle-Droplet Interactions

When a solid particle (sphere) impacts a liquid droplet…

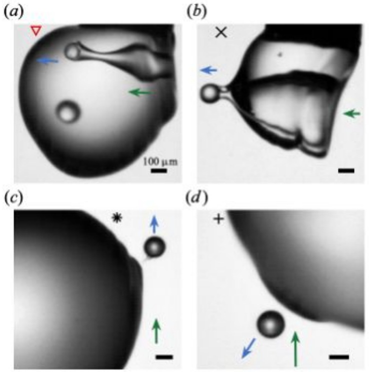
- Capture
- Rebound
- Cavity formation
- Similar mechanisms to flat interfacial interaction

Controlling Parameters

- Particle entry velocity, size, material properties, wettability
- Liquid properties, ambient gas properties
- …more?



N. B. Speirs et al. (2023)



N. B. Speirs et al. (2023)

Implications

- Raindrop particulate pollutant capture & atmospheric cleansing
- Wet scrubber systems for gaseous control

Launcher Use Cases

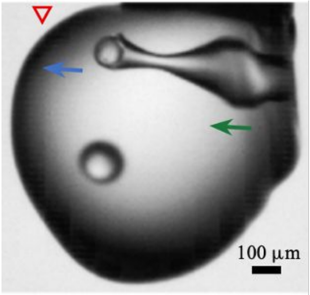
Particle-Planar Interface Impact Experiments

- Millimetric particle impacts planar liquid-air interface at prescribed velocity
- Open air or vacuum

C. Duez et al. (2007).

Particle Droplet Impact Experiments

- Millimetric particle impacts pendant droplet at prescribed velocity
- Open air or vacuum



N. B. Speirs et al. (2023).

Design & Performance Requirements

No.	Requirement	Explanation
1	**Confined space operation**	***244 x 244 x 310 mm test section (vacuum chamber), inaccessible***
2	**Particle storage & reloading**	***Store & reload particles within launcher***
3	Universal mounting	*Easy mounting scheme*
4	**Particle size**	***1-2 mm diameter***
5	Particle material compatibility	*Varying materials, densities, etc.*
6	Vacuum operation	*Vacuum chamber*
7	Splash resistance	*Splash from fluid bath/droplet*
8	**Particle velocity**	***0.9-5 m/s ±0.25 m/s***
9	**Particle trajectory**	***≤0.5 mm deviation @ 5 mm travel***
10	**Launch angles**	***-90-90˚***
11	Affordability & accessibility	*Affordable, accessible, well-documented*

Launch Velocity Model

Energy Balance Basis: Launch velocity as function of spring compression

$$v_{launch}(x_1) = \begin{cases} \sqrt{\frac{k}{m}x_1^2 + 2gd_0cos\theta - \frac{F_{nc}d_0}{m}} & x_1 \le d_0 \\ \sqrt{\frac{k}{m}(x_1^2 - (x_1 - d_0)^2) + 2gd_0cos\theta - \frac{F_{nc}d_0}{m}} & x_1 > d_0 \end{cases}$$

$k = spring\ constant$

$m = mass\ of\ plunger + sphere + spring$

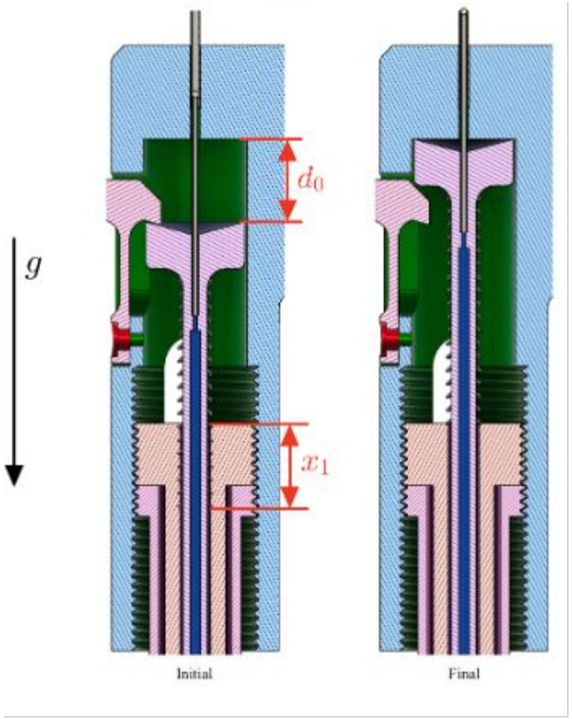
$x_1 = spring\ compression\ distance$

$g = gravitational\ constant$

$d_0 = plunger\ displacement\ distance$

$F_{nc} = net\ non - conservative\ forces$

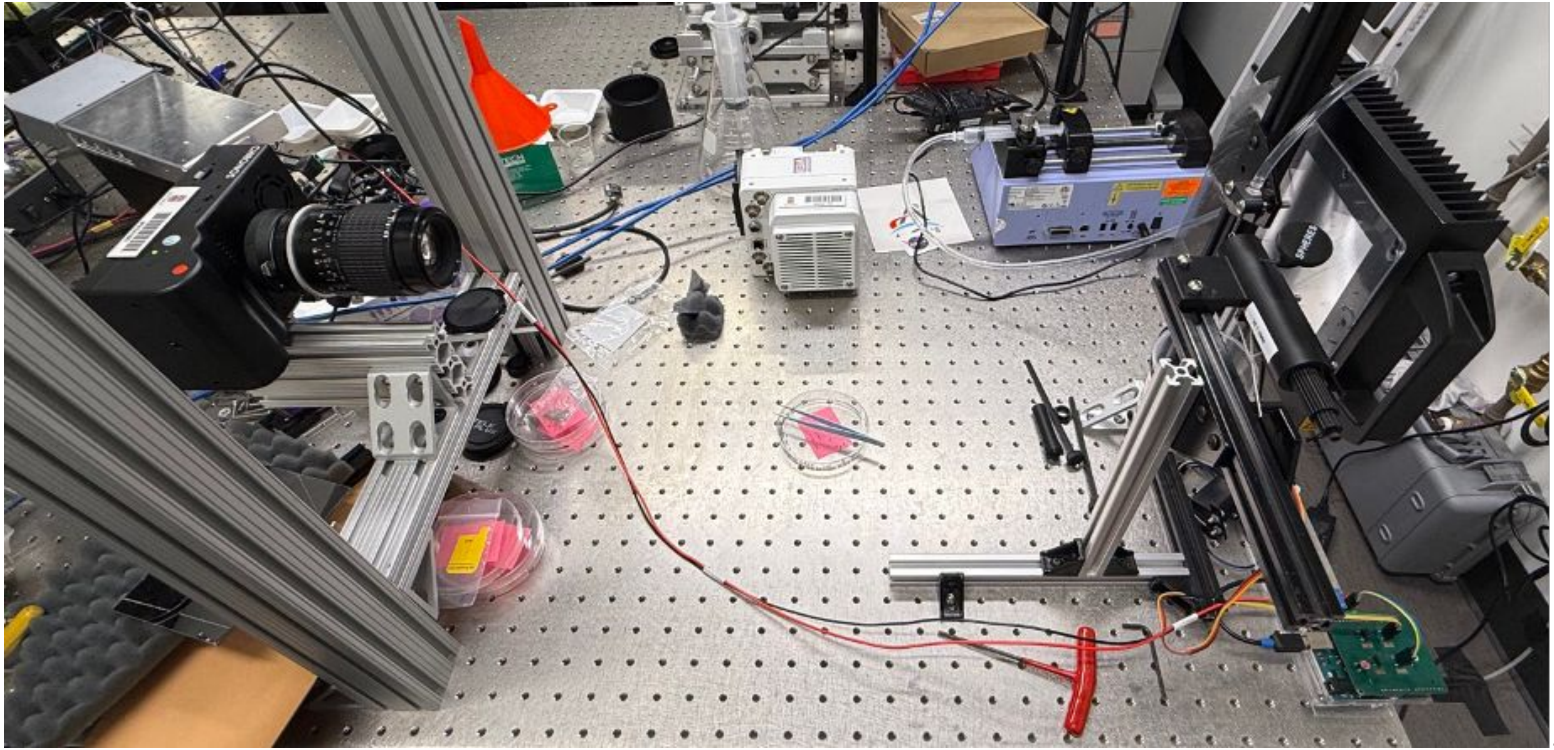
$\theta = launch\ angle\ (where\ 0^\circ\ is\ straight\ down)$



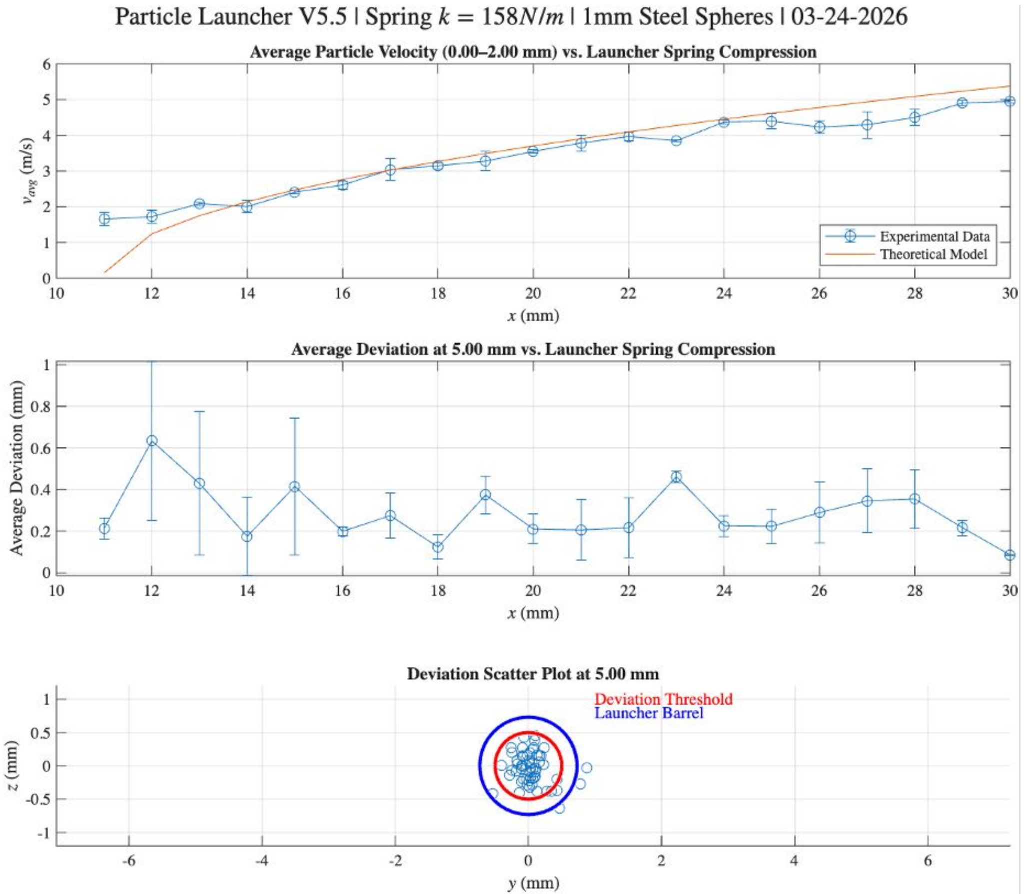
Launch angle of 180°

Performance Evaluation Methodology

Apparatus



Version 5 Performance Results



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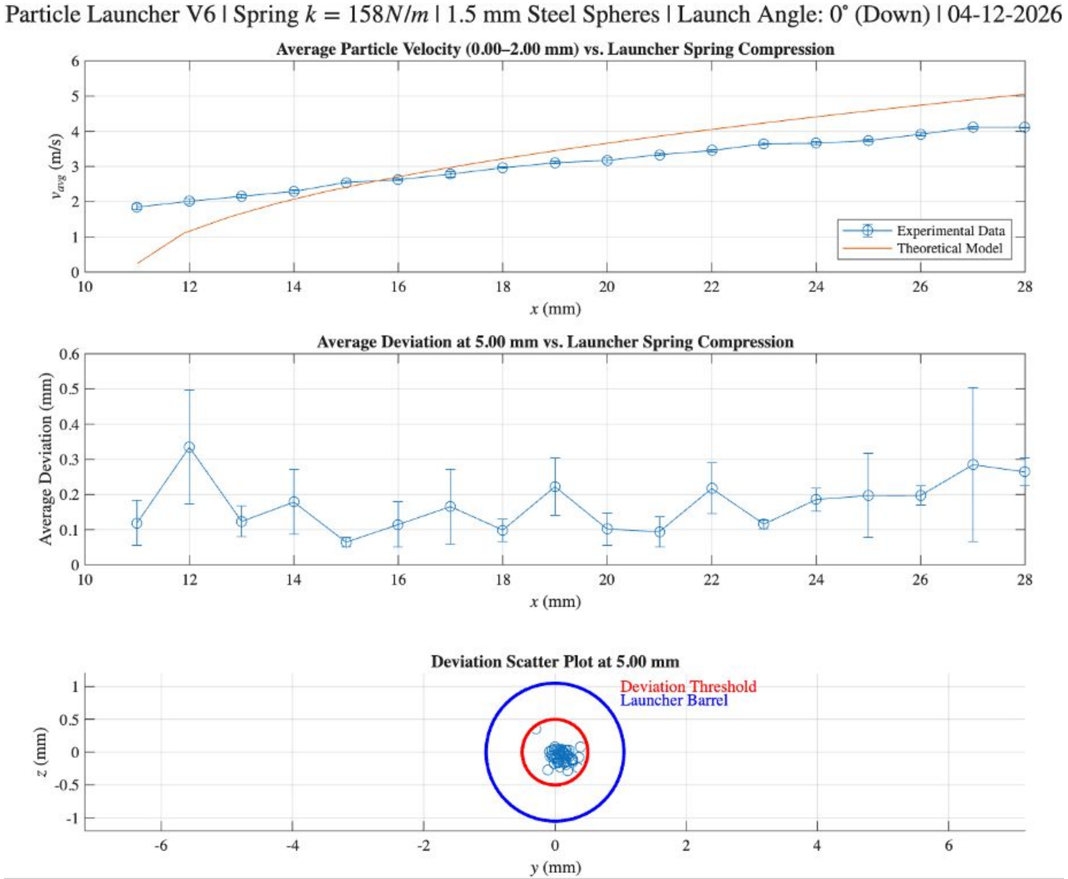
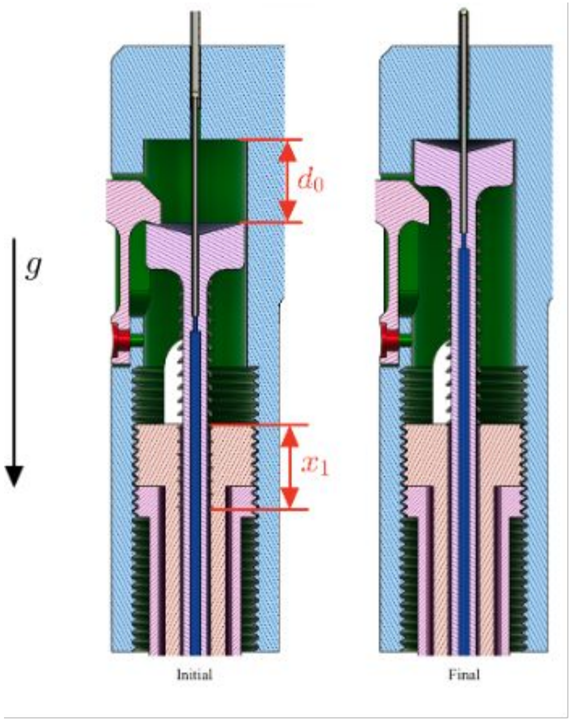
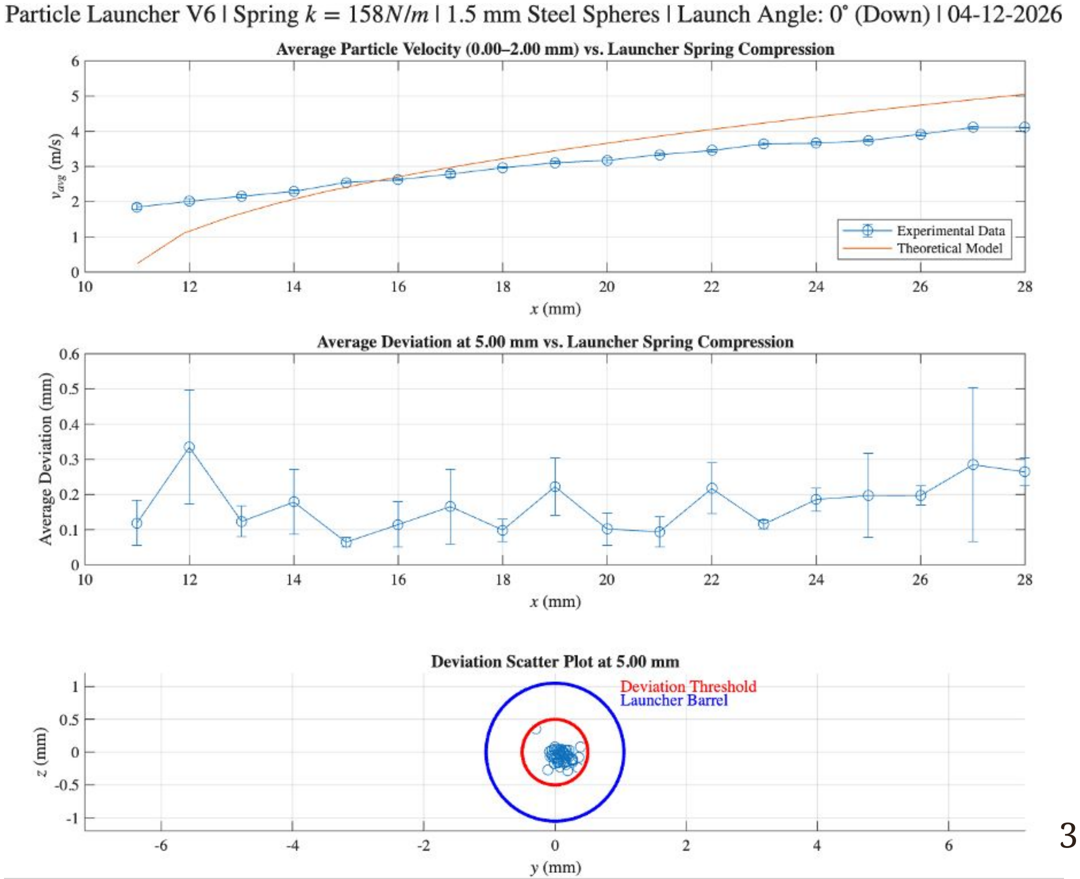
Version 6 Performance Results



Version 6 Performance Results





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Fulfillment of Design Requirements

No.	Requirement	Fulfilled?	Explanation
1	Confined space operation	*Yes*	*Fits in section, electronic operation*
2	Particle storage & reloading	*Yes*	*Storage & semi-automatic firing*
3	Universal mounting	*Yes*	*Bracket*
4	Particle size	*Yes*	*1, 1.5 mm spheres*
5	Particle material compatibility	*Yes*	*Steel, teflon*
6	Vacuum operation	*Unverified*	*Not tested under vacuum*
7	Splash resistance	*Yes*	*Operates similarly when wet*
8	Particle velocity	*Yes*	*Spans range with acceptable uncertainty*
9	Particle trajectory	*Yes*	*Deviations within thresholds*
10	Launch angles	*Yes*	*Can fire in requisite directions*
11	Affordability & accessibility	*Yes*	*Public project repository, ~$65 launcher*

Sample Experimental Performance